COMP3821 26T3 Educational Project Proposal

# How to Prove Nothing: Keeping Secrets with Zero Knowledge

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## 1 Topic and objective

The project is designed to explain the multi-party computation in the head algorithm (MPC-in the head) to an first year computer science student in a mathematically sound and understandable manner, introducing the prerequisite knowledge in cryptography topics along the way. Our intended audience is first year university students who have done the basic introductory math and computer science courses, such as UNSW Math 1A and 1B, and have a basic understanding on discrete math and algorithms.

Our objective is to create an interactive educational website that introduces the concept of zero-knowledge proofs (ZKPs) in a clear and easy to digest method. The main focus will be building up to and explaining MPC-in the head. We will detail the steps of the protocol, how is works and illustrate it's properties, notably how it maintains zero-knowledge (ZK). We will also be supporting written explanations by using interactive demonstrations where feasible, to simulate what information is available to either party, how the prover and verifier interact, and how calculations are performed.

## 2 Initial background and understanding

Our groups had little prior knowledge in cryptography prior this project. Hence we performed thorough research on ZKPs to obtain sufficient preliminary knowledge is required to give us a clear plan in fulfilling our objective, especially with the target audience of first and second years in mind. We consulted numerous sources in various forms such as websites and research publications. During this preliminary stage, we researched several ZKP procedures such as Groth16 [4], MPC-in the head [10], VOLE-in the head [1], as well as related cryptographic algorithms.

Notably, MPC-in the head [10] caught our attention as a good potential algorithm to explain in an educational project for several reasons:

1. The toy model of MPC-in the head found in [3] was explainable between peers in our group unfamiliar with cryptography concepts within an hour, deeming it feasible to explain and analyse in an educational setting.
2. It touches on many other topics in zero-knowledge proofs and cryptography such as arithmetic circuits, secret sharing, secure computation, and commitment schemes.
3. It serves as a good bridge to more advanced proof schemes, such as vector oblivious linear evaluation in the head (VOLE-in the head) [1], and zk-SNARKs [8].

Alongside formal papers, our group also investigated several educational materials and articles on cryptography. While researching Groth16 to assess it's viability as an algorithm to explain, we found A. Tomescu's website [9] explaining a broad range of ideas in cryptography, an educational article [3] explaining a basic model of multi party computation-in the head. We also found an interactive demonstration of a 3-colorability ZKP [11] and a youtube channel with a series of videos on the mathematics behind ZK [2]. These educational materials, notably the websites informed our decisionmaking process for the format we decided on and can serve as good references for our own project.

We found numerous sources on MPC-in the head which we will research ranging from the original paper [6], lecture slides [10], and other papers and theses on the topic [7]. Such investigations will also involve familiarising ourselves with commitment schemes, digital signatures and the Fiat-Shamir heuristic among other cryptography concepts found in our previous sources.

# 3 Format, scope and intended outcomes

## 3.1 Scope: Topic List

**Idea behind zero knowledge proofs:** We establish the general concept behind proofs of knowledge and ZKPs in particular. We introduce them as protocols between a prover and verifier with simple but crude examples, and discuss key properties of completeness, soundness and zero knowledge. Optionally, we explain the example of a graph 3-coloring ZKP if time and length permits. [5]

**Motivation and cryptography background:** Before explaining our particular ZKP, we motivate and establish the relevant cryptography ideas, introducing digital signature schemes, and foreshadow the Fiat-Shamir transformation to motivate ZKPs as tools to construct signatures. We also discuss commitment schemes as tools used in cryptographic protocols. If time and length permits, we can investigate and establish the exclusivity of a commitment's perfect binding and hiding.

**MPC-in the head:** The main section of the article will involve an explanation of multi-party computation (MPC) and the associated ZKP called MPC-in the head. Notably after an explanation of the protocol and ZKP we will explore characteristics such as the notion of consistency, $t$-privacy and it's relation to zero-knowledge. As an further goal, we would like to create an interactive demonstration of steps in an MPC to aid viewer understanding.

**Fiat-Shamir transformations:** We discuss the public coin nature of MPC-in the head, and explain how the Fiat-Shamir transformation can convert such interactive proofs of knowledge into digital signatures.

## 3.2 Format: Educational Website

The website will organise the topics in section 3.1 into a series of pages that are intended to be completed in order. A navigation bar will allow for easy jumping between topics and will include a progress indicator. A glossary will also be available to allow the user to search for unfamiliar terms. The website will be hosted on Github pages and will use HTML, CSS and JavaScript.

Each section will have an in depth written explanation with diagrams and examples along with an interactive demonstration of the concept it is explaining. The simulations will allow users to change inputs and follow the process to see how it affects the final result.

# 4 Proposed approaches and evaluations

Aside from the little knowledge we currently have on cryptography, another challenging aspect of our project is the fact that none of us have any experience in creating a website. This makes our aspiration to create a visual and interactive is quite an ambitious endeavour. To alleviate the workload, we decided to use GitHub pages to host our website so that we do not have to worry about the network side of the deployment.

Combining the complications due to lack of familiarity with ZKPs and building a website, it will be very difficult for us to deal with a sophisticated zero-knowledge proof system like Groth16 for example, especially due to the absence of opportunities to present this system visually and interactive. It would also be hard to not only understand the concepts behind Groth16, but also to transform our understanding into a form that other can easily understand due to the complex nature of this system.

Instead of explaining one advanced concept in depth, we agreed that it would be more beneficial and realistic to break up our website into four parts, including an introductory section, with the idea of having a progressive format in our educational material in order to work towards an understanding of a relatively simpler algorithm (MPC-in the head). This strategy not only assists us in reaching our objective within the 8 weeks but it is also suitable for our target audience who will need to develop a strong cryptography foundation before they explore more about ZKPs and the areas within it.

To determine the quality of our website and its educational ability, we aim to share the website while it is beta with people who do not have not have a strong technical background and provide a

feedback form for them to complete. This method gives us a rough idea on the pacing, readability and engagement of our website and we will use the user feedback to improve our website before the completion of our project.

Should we realise during the progress of our project that designing a website is not feasible to complete our project in a timely manner, we have also considered delivering our project in the form of a learning material for a standard university tutorial like a slideshow or video. We would aim to retain the visual aspects to demonstrate our findings, however we would have to forego the interactibility of our product to be able to complete it with fewer technical components.

# 5 Plan and collaboration

## 5.1 Project Timeline

***Week 4*** **Setup & Research:** Deploy a basic viewable website. Research topics to plan topic progression and identify key points for further research.

***Week 5*** **Static Prototype & Planning:** Create landing page and site navigation. Draft and plan the first introductory topic motivating ZKP with draft explanation and interactive parts.

***Week 6*** **Intro Page Completion:** Finalise introductory ZKP page with, including working interactive or animated elements. Draft explanations for the remaining topics.

***Week 7*** **Progress Review Demo:** Test website stability and animations. Start working on the next topic page for a functional progress demo. Finalise text components of the rest of the topics including planning for the remaining the interactive elements.

***Week 8*** **Alpha Website:** Complete all topic pages for a full run through with only some interactive or animated elements left unfinished. Verify with mentor the contents of the website for correctness and clarity of scope.

***Week 9*** **Testing & Feedback:** Finish the remaining interactive elements. Carry out user testing for feedback to refine explanations and website layout. Begin work on the final poster and report.

***Week 10*** **Final Details:** Polish and finalise website layout and explanations near the beginning of the week. Complete the poster and final report.

## 5.2 Project Management

| Collaborative Goal | Planned Approach/Strategy |
|---|---|
| Work Allocation | Work will be distributed using a kanban board on Github Projects and Issues. Work will be distributed to people's skills and availability. |
| Contribution Review | For website contributions, we will use Github pull requests. All the potential explanations for topics will be reviewed as a team weekly. |
| Shared Understanding | A weekly meeting on Wednesdays after the mentor meetup will be held to discuss concerns with the project and also refine on the draft explanations. |
| Next Steps | A shared Github repository needs to be made. Further research into the chosen topics will be completed. A basic connection to a website, likely using Github Pages will be setup. |
| Current Questions | Are there visual ideas or topics you'd like to see? Are there specific subtopics you think we should cover? Are there any concerns with the topics chosen? |

## References


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[11] E. Yang. Interactive zero knowledge 3-colorability demonstration. `https://web.mit.edu/~ezyang/Public/graph/svg.html`.